

Scalability & Future Plan

Date	13-07-2026
Team ID	
Project Name	Automated Credit Card Approval Prediction using Machine Learning
Maximum Marks	1 Mark

Scalability & Future Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Prediction depends on a static historical dataset.	The model cannot learn from newly approved or rejected applications automatically.
2	Data Storage	Free-tier Render hosting may experience cold starts.	The first request after inactivity may take a few seconds before the application becomes responsive

3	Performance	No real-time bank or credit bureau integration.	Predictions are based only on user-entered information and not on live credit history or banking records.
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This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Prediction depends on a static historical dataset.	The model cannot learn from newly approved or rejected applications automatically.	Medium
2	Free-tier Render hosting may experience cold starts.	The first request after inactivity may take a few seconds before the application becomes responsive.	Low
3	No real-time bank or credit bureau integration.	Predictions are based only on user-entered information and not on live credit history or banking records.	High

Scalability Plan:



Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Integrate Credit Score APIs and Banking APIs for real-time applicant verification.	2-3 months	Improves prediction accuracy by using live financial and credit history data.
Phase 3	Add OCR-based document verification, fraud detection, and email/SMS notification services.	3-6 months	Reduces manual verification effort, improves security, and enhances the user experience.
Phase 4	Develop an Android/iOS mobile application, multilingual support, Explainable AI (XAI), and cloud-based analytics dashboard.	6-12 months	Increases accessibility, transparency, and scalability for banks and financial institutions.